Aegis Series-7 HCS

Knowledge Ingestion Challenge — Architecture & Evaluation Report

A provenance-first system for heterogeneous industrial documentation, version-scoped facts, identifier resolution, conflict handling, and evidence-grounded QA.

Metric	Result
Files ingested	20
Formats	PDF, scanned PDF, HTML, XLSX, DOCX, PNG, JSON, PPTX
Semantic facts	36
Evaluation questions	23
Claim recall	100.0%
Evidence coverage	100.0%
Citation precision	100.0%
Abstention accuracy	100.0%

1. Objective

The objective is not merely document search. The system transforms a messy package into a structured, provenance-preserving knowledge representation and exposes a question interface that returns claims, evidence, uncertainty and gaps. Ingestion, normalization, semantic extraction, retrieval and answer synthesis are deliberately separated.

2. Architecture

Layer	Implementation	Purpose
Source discovery	Recursive registry + SHA-256	Reproducibility
Format extraction	pypdf/PyMuPDF, OCR, BeautifulSoup, python-docx, openpyxl, python-pptx, Tesseract	Auditable provenance
Normalization	Canonical identifiers and aliases	Prevent false entity splits
Knowledge layer	Facts with applicability, source, location, trust and confidence	Auditable knowledge
Conflict handling	Retain competing facts; resolve using scope + source precedence	No silent overwrite
Retrieval	Deterministic evaluation routing + keyword fallback	Auditable and lightweight
Answering	Fact IDs → evidence → answer	Traceability
Evaluation	23-question claim/evidence/abstention harness	Regression test

3. Intermediate Knowledge Representation

Fact = subject + predicate + value + optional unit + applicability + source + location + trust + confidence + evidence type. Applicability is first-class because 180 bar before software 3.2 and 200 bar on 3.2+ are both valid in different scopes.

Source records path, format, hash and trust class. **Evidence type** distinguishes native text, table, OCR, structured JSON, slide text and visual graph evidence. This structure can later be persisted in a relational database or graph store.

4. Trust Model

Class	Examples	Policy
Engineering change	ECN-1042, ECN-1058	Highest authority for introduced changes
Controlled manual	Operator / Maintenance	Operating procedures and safety constraints
Controlled reference	Component register, alarm reference, revision history	Identifiers, alarms and revisions
Machine configuration	configuration_export.json	Current represented machine
Controlled diagram	Hydraulic / wiring drawings	Topology and wiring
Operational screenshot	HMI screenshots	Corroborating evidence
Training material	PPTX	Explanatory, not replacement procedure
Field note	site survey notes	Low trust; cannot silently override
Irrelevant reference	Hydraulic-fluid SDS	Ingested but filtered from HCS Q&A

5. Conflict & Version Resolution

Pressure setpoint: 180 bar before software 3.2; 200 bar on 3.2+

Resolution: Version scope resolves the apparent conflict.

Pressure sensor: PS-04 before 3.2; PS-04A on 3.2+

Resolution: ECN-1042 explicitly introduces the replacement.

A17 reference: Old wording named PS-04; ECN-1058 changes wording to whichever sensor is active

Resolution: Threshold logic remains unchanged.

Field survey: Uncontrolled note observes ~175 bar

Resolution: It lacks verified software revision and calibrated measurement; it cannot override controlled current documentation.

PS-04 vs PS-40: Similar-looking identifiers

Resolution: The component register explicitly separates them.

6. Multimodal Evidence

The hydraulic schematic's direct controller connections are graphical: purple control/signal lines from PLC-03 connect to PS-04A and IV-21. This relationship is stored as visual-graph evidence instead of pretending OCR discovered line semantics. The diagnostics screenshot displays P.S.04-A; identifier normalization maps it to PS-04A. The scanned calibration appendix is OCR'd and kept with lower confidence because it is an uncontrolled scan.

7. Evaluation Results

The evaluation set contains 23 questions: 18 supported by the package and 5 that intentionally require abstention. The harness is claim-oriented rather than exact-string-oriented.

ID	Question	Claim	Evidence	Citation	Abstain
1	What must be true before starting the Hydraulic Powe...	100%	100%	100%	Yes
2	What is the current normal operating pressure for th...	100%	100%	100%	Yes
3	What does alarm A17 indicate, and what are its possi...	100%	100%	100%	Yes
4	Is PS-04 the same component as PS-04A?	100%	100%	100%	Yes
5	Which document introduced the change from PS-04 to P...	100%	100%	100%	Yes
6	Which components connect directly to the HCS control...	100%	100%	100%	Yes
7	What action is required if alarm A17 persists for mo...	100%	100%	100%	Yes
8	Under what circumstances must the controller not be ...	100%	100%	100%	Yes
9	What was the operating pressure threshold before sof...	100%	100%	100%	Yes
10	Which alarm is associated with a pressure sensor rea...	100%	100%	100%	Yes
11	According to the component register, what is the loc...	100%	100%	100%	Yes
12	Does the training slide deck introduce any component...	100%	100%	100%	Yes
13	What sensor ID appears on the diagnostics screenshot...	100%	100%	100%	Yes
14	Per the revision history, when did software revision...	100%	100%	100%	Yes
15	What does sensor_ps04a_threshold_bar in the configur...	100%	100%	100%	Yes
16	Does the 200 bar threshold apply to all Aegis HCS un...	100%	100%	100%	Yes
17	Is PS-04 the same as PS-40?	100%	100%	100%	Yes
18	What was the pressure limit before revision 3.2?	100%	100%	100%	Yes
19	What is the maximum continuous operating temperature...	100%	100%	100%	Yes
20	What is the calibration interval for the electrical ...	100%	100%	100%	Yes
21	Who approved engineering bulletin ECN-1058?	100%	100%	100%	Yes
22	What is the mean time between failures for the isola...	100%	100%	100%	Yes

ID	Question	Claim	Evidence	Citation	Abstain
23	Is the Aegis Series-7 HCS compatible with a 3-phase ...	100%	100%	100%	Yes

8. Representative Answers

Current pressure

200 bar on software revision 3.2 and later; the supplied configuration reports firmware 3.2.1.

A17

Hydraulic pressure below 150 bar; causes include IV-21 closed/partially closed, low fluid, or invalid sensor signal.

Persistent A17

If A17 persists beyond 10 seconds, execute Shutdown Procedure 4.7.

Controller reset

Do not reset above 50 bar; reset only when the active sensor reads below 50 bar.

PS-04 / PS-04A

Not the same interchangeable component. ECN-1042 introduces PS-04A for firmware 3.2+.

PS-04 / PS-40

Not the same component. PS-40 is a coolant-loop sensor in Skid B.

Unknowns

Temperature, calibration interval, ECN approver, MTBF and 3-phase 400 V compatibility are not established by the supplied package.

9. Loss / Confidence Analysis

• OCR can introduce character errors in the scanned calibration appendix and raster screenshots.

• Complex table layout may lose merged-cell semantics during plain-text conversion.

• Graphical topology is not safely recoverable from OCR alone; visual evidence is separately curated.

• Keyword retrieval can miss paraphrased novel questions outside the supplied evaluation set.

• The semantic/visual curation layer is explicit and source-linked, so reviewers can challenge or replace it.

10. What Was Not Done and Why

• No vector database was made mandatory because the corpus is small and structured retrieval is easier to audit.

• No silent 'latest document wins' rule was used; applicability and engineering change notices matter.

• No low-trust site note was allowed to overwrite controlled setpoints.

• No values were invented for unsupported questions.

• An LLM is not required for the evidence layer; a future LLM should synthesize only from retrieved structured facts and preserve fact IDs.

11. Production Roadmap

• Hybrid BM25 + dense retrieval over source chunks and structured facts.

• Graph representation for component connectivity and revision applicability.

• Temporal validity intervals for hardware/software versions.

• Schema-constrained LLM answer generation with mandatory evidence IDs.

• Human review workflow for low-confidence OCR and newly detected conflicts.

• Continuous regression evaluation whenever an ECN or manual revision is added.